

Subir Majumder PhD

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- Highlights**
- Researcher passionate about energy systems optimization, computing and learning, and economics.
 - Demonstrated ability to conduct original research and develop novel approaches.
 - Work individually or lead a multi-disciplinary, remotely-connected team.
- Research Interests**
- Applied Power System Economics with Behaviour Analysis.
 - (Learning) Algorithms for Power & Energy Systems.
 - Power System Modeling and Planning to Achieve Sustainable Electrification.
 - Applied Cyber-Physical Energy System Resiliency.
- Education**
- 2020, Ph.D., Indian Institute of Technology Bombay (IITB), India and University of Wollongong (UOW), Australia (under a Cotutelle/Joint Agreement).**
- Dissertation:** *Techno-economic analysis of electricity networks with renewable energy sources and storage devices.*
- Principal Advisors: Dr. S. A. Khaparde and Dr. A. P. Agalgaonkar
- *The dissertation has been presented in a panel session by Dr. S. V. Kulkarni at the IEEE Power and Energy Society General Meeting, 2019.*
- 2020, M.Tech. in Energy Systems Engineering (under M.Tech. + Ph.D. Dual Degree program), Indian Institute of Technology Bombay, India.**
- 2012, B.Tech. in Electrical Engineering, Kalyani Govt. Engineering College, West Bengal University of Technology, India.**
- Research Experience**
- November 2024 - Present, Post-Doctoral Fellow, Harvard University, MA, USA.**
- Developing data-driven behavioral model AI and cryptocurrency mining of data centers. This position is jointly with *Texas A&M University*. (Mentor: Dr. Le Xie).
- April 2023 - Present, TEES Senior Research Engineer I, Texas A&M University, TX, USA.**
- Contributed to two projects: US DOE Dynamic Watermarking and US DOE Open COG-Grid. Research in cyber-physical system security, demand response, electricity markets, distributed optimization, and stochastic power system control considering Distributed Energy Resource (DER) systems. Assisted in training masters and Ph.D. students to apply electrical engineering fundamentals. (Mentor(s): Dr. Le Xie, Dr. I. Aravena).
- September 2021 - March 2023, Engineering Scientist, West Virginia University, WV, USA.**
- Contributed to two projects: US DOE UI-ASSIST and NSF CPS aDaption. Research in distributed optimization, resilient operation of the power distribution system, and development of retail electricity markets with Distributed Energy Resource (DER) systems. Mentoring PhD students for distributed algorithm development with discrete variables and test-bed development for distributed applications. Collaborated for machine learning-based power system operational control algorithm development. Contributed to writing proposals as a researcher or a co-I of PSERC, NSF, and US-DOE projects. (Mentor: Dr. A. Srivastava).
- January 2020 - August 2021, Post-Doctoral Research Associate, Washington State University, WA, USA.**
- Research in distributed optimization and resilient operation of the power distribution system with DERs. Mentoring undergraduate students for test-bed development.

Collaborated for machine learning-based power system operational control algorithm development. Contributed to writing proposals for external funding. (Mentor: Dr. A. Srivastava).

July 2019 - December 2019, Research Associate, Indian Institute of Technology Bombay, Mumbai, India.

Research in power quality economics. (Mentor: Dr. S. A. Khaparde).

Journals

- [J16] I. Aravena, C.-C. Sun, R. Shi, **S. Majumder**, W. Yan, J.-Y. Joo, L Xie and J. Wang, "Open Power System Datasets and Open Simulation Engines: A Survey Towards Machine Learning Applications," *IEEE Open Access Journal of Power and Energy*, vol. 12, pp. 353-365, 2025.
- [J15] L. Xie, **S. Majumder**, T. Huang, Q. Zhang, P. Chang, D. J. Hill, M. Shahidehpour, "The Role of Electric Grid Research in Addressing Climate Change," *Nat. Clim. Chang.*, vol. 14, pp. 909-915, 2024.
- [J14] **S. Majumder***, L. Dong*, F. Doudi*, Y. Cai*, C. Tian, D. Kalathil, K. Ding, A.A. Thatte, and L. Xie, "Exploring the Capabilities and Limitations of Large Language Models in the Electric Energy Sector," *Joule*, vol. 8, 1-6, 2024. (* → equal contribution).
- [J13] **S. Majumder**, S. A. Khaparde, A. P. Agalgaonkar, S. V. Kulkarni, A. K. Srivastava, and S. Perera, "Chance-Constrained Pre-Contingency Joint Self-Scheduling of Energy and Reserve in a VPP," *IEEE Trans. Power Syst.*, vol. 39, no. 1, Jan. 2024, pp. 245-260 (also presented in *IEEE PESGM*, Jul. 2024).
- [J12] **S. Majumder**, A. Vosughi, H. M. Mustafa, T. E. Warner, and A. K. Srivastava, "On the Cyber-Physical Needs of DER-based Voltage Control Algorithms," *IEEE Access*, vol. 11, 2023, pp. 64397-64429.
- [J11] S. U. Kadir, **S. Majumder**, A. K. Srivastava, A. Chhokra, A. Dubey, H. Neema, and A. Laszka, "Reinforcement Learning based Proactive Control for Transmission Grid Resilience to Wildfire," *IEEE Trans. Indus. Inform.*, vol. 20, no. 1, Jan. 2024, pp. 795-805.
- [J10] **S. Majumder**, N. Patari, A. K. Srivastava, P. Srivastava, and A. M. Anaswamy, "Epistemology of Voltage Control in DER-Rich Power System," *Elec. Pow. Syst. Res.*, Art. No. 108874, 2023.
- [J9] **S. Majumder**, "Premium Power Investment Strategy utilizing the Economy of Scale of Custom Power Devices," *Elec. Pow. Syst. Res.*, Art. No. 108743, 2023.
- [J8] **S. Majumder**, G. Kandaperumal, S. Pandey, A. K. Srivastava, and C. Koplin, "Pre-Event Two-Stage Proactive Control for Enhanced Resiliency," *IEEE Access*, vol. 10, pp. 83281-83296, 2022.
- [J7] A. Vosughi, A. Tamimi, A. B. King, **S. Majumder**, and A. K. Srivastava, "Cyber-physical Vulnerability and Resiliency Analysis for DER Integration: A Review, Challenges and Research Needs," *Renew. Sust. Energ. Rev.*, vol. 168, Art. no. 112794, 2022.
- [J6] **S. Majumder**, S. A. Khaparde, A. P. Agalgaonkar, S. V. Kulkarni, and S. Perera, "Graph Theory Based Voltage Sag Mitigation Cluster Formation Utilizing Dynamic Voltage Restorers in Radial Distribution Networks," *IEEE Trans. Power Del.*, vol. 37, no. 1, pp. 18-28, Dec. 2020.
- [J5] **S. Majumder**, A. P. Agalgaonkar, S. A. Khaparde, S. Perera, S. V. Kulkarni, and P. P. Ciufu, "Allowable Delay Heuristic in Provision of Primary Frequency Reserve in the Future Power System," *IEEE Trans. Power Syst.*, vol. 35, no. 2, pp. 1231-1241, Aug. 2019.
- [J4] **S. Majumder**, A. P. Agalgaonkar, S. A. Khaparde, P. P. Ciufu, S. Perera, and S. V. Kulkarni, "Allocation of Common-Pool Resources in an Unmonitored Open

System,” *IEEE Trans. Power Syst.*, vol. 34, no. 5, pp. 3912-3920, Sept. 2019.

[J3] **S. Majumder**, S. A. Khaparde, A. P. Agalgaonkar, P. P. Ciufu, S. Perera, and S. V. Kulkarni, “DFT-Based Sizing of Battery Storage Devices to Determine Day-Ahead Minimum Variability Injection Dispatch with Renewable Energy Resources,” *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 626-638, Jan. 2019, (also presented in *IEEE PESGM*, Aug. 2018).

[J2] **S. Majumder**, R. M. Shereef, and S. A. Khaparde, “Two-Stage Algorithm for Efficient Transmission Expansion Planning with Renewable Energy Resources,” *IET Renew. Power Gen.*, vol. 11, no. 3, pp. 320-329, Feb. 2017.

[J1] **S. Majumder** and S. A. Khaparde, “Revenue and Ancillary Benefit Maximisation of Multiple Non-collocated Wind Power Producers Considering Uncertainties,” *IET Gen. Transm. Distrib.*, vol. 10, no. 3, pp. 789-797, Jul. 2016.

Newsletters & Book Chapter

[B1] G. Kandaperumal, **S. Majumder**, and A. K. Srivastava, “Microgrids as a Resilience Resource in the Electric Distribution System,” *book chapter in Elsevier-Electric Power Systems Resiliency – Modelling, Opportunity and Challenges*, Paperback ISBN: 9780323855365, 1st Ed., July 1, 2022.

[N2] **S. Majumder**, and A. K. Srivastava, “Resilience-Driven Integration of Distributed Energy Resource (DER): Coordinating DER Services for Value,” in *IEEE Smart Grid Newsletters*, Nov. 2022.

[N1] **S. Majumder**, and A. K. Srivastava, “Resilience-driven Integration of Distributed Energy Resource (DER): Holistic Value Analysis,” in *IEEE Smart Grid Newsletters*, Sept. 2022.

Conference Publications

[C13] **S. Majumder**, I. Aravena, L. Xie, “An Econometric Analysis of Large Flexible Cryptocurrency-mining Consumers in Electricity Markets,” accepted for publication in HICCS 2025, (Available: https://sum-em.github.io/files/LFL_SARIMA_HICCS_2024-1.pdf).

[C12] M. Zeid, **S. Majumder**, H. Ibrahim, P. Enjeti, L. Xie, C. Tian, “Predicting DC-Link Capacitor Voltage and Current Ripple in AC-DC Rectifier Circuits Using Fine-Tuned Large Language Models,” accepted for publication in IECON 2024, (Available: <https://arxiv.org/abs/2407.01724>).

[C11] A. Samanta, **S. Majumder**, H. Ibrahim, P. Enjeti, and L. Xie, “EMT Model of Cryptocurrency Mining Loads to analyze their LVRT Capability,” in *IEEE PESGM 2024*.

[C10] **S. Majumder**, P. Fu, and L. Xie, “A mechanism to account for locational carbon impact in electricity markets,” in *Texas A&M Conference on Energy 2023*, Sept. 2023.

[C9] D. Kim, **S. Majumder**, and L. Xie, “Line-Post Insulator Fault Classification Model using Deep Convolutional GAN-based Synthetic Images,” in *NAPS*, 2023.

[C8] P. S. Sarker, **S. Majumder**, M. F. Ruffy, and A. K. Srivastava, “Impact Analysis of Cyber-Events on Distributed Voltage Control with Active Power Curtailment,” in *PEDES*, 2022.

[C7] J. Adan, **S. Majumder**, and A. K. Srivastava, “Distributed Optimization Approaches with Discrete Variables in the Power Distribution Systems,” in *NAPS*, 2022.

[C6] P. S. Sarker, N. Patari, B. Ha, **S. Majumder**, and A. K. Srivastava, “Quasi-Static Cyber-Power Co-Simulation Platform for Performance Analysis of Distributed VAR Control during Cyber-Attacks,” in *MSCPES*, May 2022.

[C5] S. Knudsen, **S. Majumder**, and A. K. Srivastava, “Securely implementing and managing neighborhood solar with storage and peer to peer transactive energy,” in *Paris Session – CIGRE*, Aug. 2022.

[C4] **S. Majumder**, S. A. Khaparde, V. Pradhan, A. P. Agalgaonkar, and S. Perera,

	“Optimal Voltage Sag Mitigation Solution Provision using Customers Approximate Marginal Willingness-to-Pay Function,” in <i>IEEE PowerTech</i>, Jun. 2021. [C3] S. Majumder, S. A. Khaparde, A. P. Agalgaonkar, S. V. Kulkarni, S. Perera, and P. P. Ciufo, “Hybrid Storage System Sizing for Minimum Daily Variability Injection,” in <i>TENSYMP</i>, Jun. 2019. [C2] S. Majumder, S. A. Khaparde, V. Pradhan, S. V. Kulkarni, A. P. Agalgaonkar, S. Perera, and P. P. Ciufo, “A Critical Review on Methods for Calculating the Risk of Process Failure because of Voltage Sags,” in <i>POWERCON</i>, Oct. 2016. [C1] N. V. Vigil, S. Majumder, V. Pradhan, Padmini V., and S. A. Khaparde, “Transmission Expansion Planning with Variable Wind Source,” in <i>NPSC</i>, Dec. 2014.
Manuscripts Under Review / Preparation	[R2] D. Kim, S. Majumder, and L. Xie, “A Multi-source Data Repository and Profit-robust Framework for Energy Storage Planning,” under review in NAPS, (Repository: https://sum-em.github.io/files/energy-storage-2024_Fall__IEEE_PES_GM_2025-1.pdf). [R1] S. Majumder, X. Chen, and L. Xie, “Filter-Based Zeroth-Order Methods for Voltage Control in Distribution Grids: A Practical Approach,” under first revision in <i>IEEE Trans. Indus. Inform.</i>, (Available: https://sum-em.github.io/files/Practical_Implementation_Zeroth_Order_Control_TII-1.pdf).
Fellowships and Awards	• Conferred POSOCO Power System Awards (PPSA) under Doctoral category, 2020. • Recommended for the Best Ph.D. Thesis award, IITB, 2020. <i>One of the selected external examiners stated: “the thesis presents the research carried out by the student in an innovative (way of thinking) manner addressing the problem of low inertia due to renewable energy integration — Application of ‘Diverse’ Approaches, Conceptually and Mathematically for Problem Solving.”</i> • Recommended for special commendation for an Outstanding Ph.D. Thesis, UOW, 2020. <i>One of the selected external examiners noted: “the thesis is a solid and addresses a number of topics that have resulted in high quality publications. In addition, he/she highlights the fact the student is able to identify research problems and develop innovative solutions.”</i> • Granted University Postgraduate Award (UPA) and Institute Postgraduate Tuition Award (IPTA) from the University of Wollongong, Australia, 2016-2017. • Granted TA-ship for M.Tech.+Ph.D. program, Govt. of India, 2012-2018. • Awarded full free scholarship from Kalyani Govt. Engineering College, Govt. of West Bengal, India on merit-cum-means basis, 2008-2012.
Teaching Experience	• Teaching Assistant + Delivered a few lectures in <i>Smart Grids</i> course at IITB, Jul.-Dec., 2018. • Teaching Lab demonstrator, <i>Power Electronics and Drives</i>, UOW, Jul.-Nov., 2017. • Teaching Lab demonstrator, <i>Power Engineering 1</i>, UOW, Feb.-Jun., 2017. • Teaching Assistant, <i>Energy Management</i>, IITB, Jan.-May, 2016. • Assisted in conducting the (4 days) CEP course on <i>Energy Management</i> at IITB, Nov., 2015 and Mar., 2015. • Head Teaching Assistant, <i>Introduction to Electrical Systems</i>, IITB, Jul.-Dec. 2014 and Jul.-Dec. 2015. • Teaching Assistant, <i>Power Systems</i>, IITB, Jan.-May 2014 and Jan.-May 2015. • Teaching Assistant, <i>Fundamentals of Energy Engineering</i>, IITB, Jan.-May 2013.
Research Mentoring Experience	• Mentored one MS student and two PhD students, May. 2023 - Jul. 2024, TAMU. (Students: A. Samanta, M. Zeid, L. Dong). • Mentored two PhD students, Jan. 2022 - Apr. 2023, WVU. (Students: J. Adan, Md. F. Rafy).

	• Supported in mentoring one undergraduate and one postgraduate student, Jan.-Dec. 2021, WSU/WVU. (Students: B. Ha, S. R. Bhavirisetty). • Supported in mentoring two undergraduate summer students, May-Aug. 2020, WSU. (Students: T. E. Warner, A. B. King). • Mentoring one undergraduate student, Feb.-Jun. 2017, UOW. (Student: M. Ayad).
Technical Skills	• Proficient in <i>GAMS</i>, <i>Gurobi-Py</i> as a mathematical modeling and optimization tool. • Proficient in <i>Matlab</i>, <i>MatPower</i>, <i>PowerWorld</i>, and <i>OpenDSS</i>. • Basic use of <i>Python</i> and <i>C</i>. • Languages: Bengali (proficient), English (proficient), German (basic, received training from DAAD) and Hindi (basic).
Invited Talks	• “Necessity of system-aware grid edge operations in the Power Grids,” organized by Pran Foundation, India, Mar. 2025. • Panelist at “The Dual Edge of Technology Panel: Equity and Sustainability in AI Usage,” organized by Harvard University IT, Boston, MA, Mar. 2025. • “Behavior of Large Cryptocurrency-Mining Firms in Electricity Markets: A Texas Econometric Analysis,” in the <i>Energy & Power Group Seminar</i>, at <i>Texas A&M University</i>, College Station, TX, Sep. 2024. • “Operational Control of the Power Grid through AI in the advent of Wildfire,” in the panel <i>Resiliency of the Power System for Sustainability</i>, at <i>2024 IEEE Power and Energy Society General Meeting</i>, Seattle, WA, Jul. 2024.
Services and Associations	• Member, <i>IEEE</i>, <i>IEEE Power and Energy Society</i>, <i>IEEE Control Systems Society</i>. • Chartered Engineer, <i>The Institute of Engineers (India)</i>. • Member, <i>American Economic Association</i>. • Member, IEEE WG on Natural Disaster Mitigation Methods and Operation Technology; Member, IEEE WG on Modern & Future Distribution System Planning. • Reviewer, <i>IEEE Trans. Power Del.</i>, <i>IEEE Access</i>, <i>IEEE Trans. Sust. Energy</i>, <i>IEEE Trans. Power Syst.</i>, <i>Energies</i>, <i>PESGM 2018</i>. • Technical Program Committee member, AIBThings conference, 2023. • Proctor, <i>IEEE xTreme 2017</i>; Judge, <i>14th Annual Summer Undergraduate Research Symposium at WVU</i>, <i>2022 Spring Undergraduate Research Symposium at WVU</i>, <i>2022 - 2024 IEEE PES GM Student Poster Competition</i>; <i>Student Mentor</i>, <i>2023 IEEE PES GM</i>, Mentor K-12 students in <i>Independent Study and Mentorship program</i>. • Mentorship and awareness initiatives at Pran Foundation, supporting marginalized students in India, 2022-present.
Professional Development	Strengthen Your Professional Communication Skills (WVU LOD, 2021), Navigating Mentoring for Undergraduate Research Mentors (WVU Workshop, 2022), Promoting Active Learning Online (ACUE microcredential Course, 2023).
Career Objectives	With over a decade of experience across three continents, I am committed to advancing power and energy systems engineering, optimization theory, and electricity markets to meet the critical challenges of sustainable, affordable, and reliable energy. My focus is on developing innovative computational algorithms and data-driven strategies that not only enhance grid reliability and integrate renewable resources but also support energy transitions. I am passionate about mentoring the next generation of engineers and researchers, preparing them to create impactful, scalable solutions that address the grand challenges of our energy future.